

RECORD

Field	Details
Course	Full Stack Web Development
Course Code	23CM4121
Experiment No.	1
Student Name	R Varshini
Roll No.	A24126552115
Date	30-08-2026

1. TITLE

Design a Static Webpage Using Pure Semantic HTML Components

2. AIM

To design a static webpage using only semantic HTML elements, without any CSS styling, and to demonstrate how browsers structure and render a document based purely on its markup.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and apply semantic HTML5 elements such as header, nav, section, main, article, aside and footer.
- To structure page layout using HTML tables for header and content columns.
- To create a user inquiry form using native form controls (text, email, select, textarea, submit and reset).
- To use lists (ordered and unordered) to present structured information.
- To observe how a browser renders a page relying entirely on default styles.

4. THEORY

Semantic HTML elements clearly describe their meaning to the browser and to the developer. Tags such as <header>, <nav>, <section>, <main>, <article>, <aside> and <footer> divide a page into meaningful regions instead of relying on generic <div> containers, which improves accessibility and helps screen readers and search engines interpret the page structure.

HTML tables (<table>, <tr>, <td>) were originally used for tabular data but can also be used for basic page layout, arranging content into rows and columns, as demonstrated here for the header bar and the two-column article/sidebar layout.

HTML forms allow users to submit data using controls such as <input type="text">, <input type="email">, <select>, and <textarea>, wrapped inside a <fieldset> with a <legend> to group related fields. The required attribute enables built-in browser-side validation without any JavaScript.

Because no CSS is used, the page relies entirely on the browser's default (user-agent) stylesheet for fonts, spacing and colors, which is why headings, links, and form controls appear in their standard unstyled form.

5. ALGORITHM / PROCEDURE

- Create a new HTML file and set up the basic document structure with <!DOCTYPE html>, <html>, <head> and <body>.

- ## 6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>

  <!-- Header Section -->
  <header>
    <table width="100%" border="0" cellspacing="0" cellpadding="10">
      <tr>
        <td>
          <h1>StaticWeb</h1>
        </td>
        <td align="right">
          <nav>
            <a href="#home">Home</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
            <a href="#articles">Articles</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
            <a href="#about">About</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
            <a href="#contact">Contact Us</a>
          </nav>
        </td>
      </tr>
    </table>
  </header>

  <hr>

  <!-- Hero / Showcase Section -->
  <section id="home">
    <center>
      <h2>Semantic Structures Without CSS</h2>
      <p>This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.</p>
      <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
    </center>
  </section>

  <hr>
```

```

<!-- Main Content Layout using a Structural Table -->
<table width="100%" border="0" cellspacing="0" cellpadding="20">
  <tr valign="top">

    <!-- Left Side: Main Content Column -->
    <td width="70%">
      <main id="articles">

        <article>
          <h2>1. Core Structural Tags</h2>
          <p>Published: July 7, 2026</p>
          <p>When you omit cascading style sheets entirely, the semantic markup
handles all of the structural heavy lifting. Elements like <code>&lt;main&gt;</code>,
<code>&lt;article&gt;</code>, and <code>&lt;section&gt;</code> tell screen readers and web
scrapers exactly how information flows.</p>
          <p>Using raw browser defaults ensures excellent loading performance,
perfect accessibility compliance, and absolute structural clarity.</p>
        </article>

        <br><hr><br>

        <article>
          <h2>2. Native Interaction Elements</h2>
          <p>Published: July 5, 2026</p>
          <p>Browsers come equipped with excellent interactive elements by default.
We can capture text inputs, provide multiple-choice configurations, and submit structured
information to backend parsers using pure markup.</p>

          <h3>Interactive Demonstration Forms</h3>
          <form id="contact" action="#" method="post">
            <fieldset>
              <legend><b>User Inquiry Form</b></legend>
              <p>
                <label for="name">Full Name: </label>
                <input type="text" id="name" name="username" required
placeholder="John Doe">
              </p>
              <p>
                <label for="email">Email Address: </label>
                <input type="email" id="email" name="useremail" required>
              </p>
              <p>
                <label for="topic">Inquiry Topic: </label>
                <select id="topic" name="usertopic">
                  <option value="general">General Feedback</option>
                  <option value="technical">Technical Architecture</option>
                  <option value="academics">Academic Curriculum</option>
                </select>
              </p>
              <p>
                <label for="message">Message Details:</label><br>
                <textarea id="message" name="usermessage" rows="5" cols="40"
placeholder="Type your notes here..."></textarea>
              </p>
              <p>
                <input type="submit" value="Submit Form Data">
                <input type="reset" value="Clear Entries">
              </p>
            </fieldset>
          </form>
        </article>

      </main>
    </td>

    <!-- Right Side: Sidebar Column -->
    <td width="30%" bgcolor="#f4f4f4">
      <aside id="about">

```

```

        <h3>Document References</h3>
        <p>Static pages map data out directly without real-time database queries or
processor overhead.</p>

        <h4>Core Benefits:</h4>
        <ul>
            <li>Instant browser painting</li>
            <li>Low memory footprints</li>
            <li>High data compatibility</li>
        </ul>

        <h4>Required Components:</h4>
        <ol>
            <li>Document Type Definition</li>
            <li>Semantic Wrapper Blocks</li>
            <li>Data Input Control Interfaces</li>
        </ol>
    </aside>
</td>

</tr>
</table>

<hr>

<!-- Footer Section -->
<footer>
    <center>
        <p>&copy; 2026 Pure HTML Architecture. Assembled completely without style
configurations.</p>
    </center>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<header>, <nav>, <footer>	Semantic elements marking the page's top banner, navigation links, and bottom footer region.
<section>, <main>, <article>, <aside>	Semantic containers dividing the page into a hero area, primary content, individual articles, and a sidebar.
<table>/<tr>/<td> (layout)	Used here purely for positioning the header items and the two-column article/sidebar layout, not for tabular data.
<fieldset> / <legend>	Groups related form controls under the "User Inquiry Form" with a visible caption.
required attribute	Triggers the browser's built-in validation, preventing form submission if the Name or Email field is empty.
<select> / <option>	Provides a dropdown of predefined inquiry topics for the user to choose from.
<textarea>	Provides a multi-line text input for the message details.
<input type="submit"> / <input type="reset">	Native buttons to submit the form data or clear all entered values.
/	Unordered and ordered lists used in the sidebar to present benefits and required components.

8. TEST CASES AND EXECUTION DATA

Each section and form control was verified by loading the page in a browser and interacting with it.

Test Case	Feature Tested	Expected Result	Status
TC-01	Page load	Header, hero section, articles, sidebar and footer render correctly	Pass
TC-02	Navigation links	Home, Articles, About and Contact Us links are visible and clickable	Pass
TC-03	Two-column layout	Articles occupy 70% width; sidebar occupies 30% with shaded background	Pass
TC-04	Form - required fields	Submitting with Name or Email empty is blocked by the browser	Pass
TC-05	Form - dropdown and textarea	Inquiry Topic can be selected; Message Details accepts multi-line text	Pass
TC-06	Reset button	Clicking Clear Entries resets all form fields to blank	Pass

9. OUTPUT

Output 1: Full Page Layout

The rendered page shows the header with the site title and navigation, the hero section, and the two-column layout with articles on the left and the sidebar on the right, all using unstyled default browser formatting.

StaticWeb

[Home](#) | [Articles](#) | [About](#) | [Contact Us](#)

Semantic Structures Without CSS

This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.

[Read Our Articles Below.](#)

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like `<main>`, `<article>`, and `<section>` tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

Document References

Static pages map data out directly without real-time database queries or processor overhead.

Core Benefits:

- Instant browser painting
- Low memory footprints
- High data compatibility

Required Components:

- Document Type Definition
- Semantic Wrapper Blocks
- Data Input Control Interfaces

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

General Feedback

Message Details:

Type your notes here...

Submit Form Data

Clear Entries

Output 2: User Inquiry Form Filled

The form fields accept user input as expected, with the Full Name, Email Address, Inquiry Topic and Message Details all editable through native browser controls.

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

Message Details:

10. RESULT

Thus, a static webpage was successfully designed using pure semantic HTML components, including header, navigation, section, main, article, aside and footer elements, a table-based layout, and a native HTML form with required-field validation. The page rendered correctly in the browser using only default styling, and all sections and form controls were verified as working as expected.